

Candlestick Visual-Regime Discovery

Code-Only Package User Manual

Using Separate Auto-Scaled and Fixed-Scale Datasets

Package	final_year_project_2_code_only.zip
Main program	run_unsupervised_regime_discovery.py
Supported datasets	Auto-scaled candlestick charts and fixed-scale candlestick charts
Execution environments	Windows and Google Colab

This manual applies to the code-only version of the project. The datasets are uploaded and stored separately.

Contents

1. Purpose of the package
2. What is included and excluded
3. Required dataset structure
4. Recommended Google Drive layout
5. Software and hardware requirements
6. Understanding the configuration files
7. Running on Google Colab
8. Running on Windows
9. What the pipeline does
10. Output folders and files
11. How to select the number of clusters
12. Comparing auto-scaled and fixed-scale experiments
13. Safe reruns and saved embeddings
14. Troubleshooting
15. Recommended experimental workflow
16. Final checklist

1. Purpose of the package

The package performs unsupervised visual-regime discovery using only candlestick chart images. It converts each candlestick image into a pretrained CNN embedding, reduces the embedding dimension with PCA, evaluates several K-Means cluster counts, recommends a practical value of K, and saves the final cluster assignments and visual inspection files.

The software is deliberately separated from the datasets. This allows the same program to be applied to two independently stored datasets:

- An auto-scaled candlestick dataset, where each chart is scaled to its own local price range.
- A fixed-scale candlestick dataset, where chart scaling follows the fixed-scale generation method.

The two datasets must be processed as separate experiments. Their outputs are written to different folders so that their cluster structures can later be compared.

2. What is included and excluded

2.1 Included in the code-only ZIP

File	Purpose
run_unsupervised_regime_discovery.py	Main CNN embedding, PCA, K-Means, reporting, and montage program.
config_autoscaled.json	Configuration template for the auto-scaled dataset.
config_fixed_scale.json	Configuration template for the fixed-scale dataset.
config.json	Generic custom configuration template.
run_on_colab.ipynb	Google Colab notebook for selecting either external dataset.
install_requirements.bat	Windows dependency installer.
run_autoscaled_windows.bat	Windows launcher for the auto-scaled dataset.
run_fixed_scale_windows.bat	Windows launcher for the fixed-scale dataset.
requirements.txt	Required Python packages.

2.2 Not included

- No candlestick images.
- No auto-scaled dataset.
- No fixed-scale dataset.
- No generated embeddings, PCA model, K-Means model, plots, or cluster reports.
- No user-specific Windows path is hardcoded in the Python source code.

3. Required dataset structure

Each external dataset must already be prepared as a candlestick-only unsupervised dataset. Both datasets must use the same internal structure.

```
dataset_folder/
├─ images/
│  ├─ sample_0_29.jpg
│  ├─ sample_1_30.jpg
│  └─ ...
└─ manifests/
   ├─ train_samples.csv
   └─ validation_samples.csv
```

The CSV manifests must contain at least the following columns:

Column	Meaning
sample_name	Example: sample_0_29
start_index	First raw-data row used by the chart window.
stop_index	Last raw-data row used by the chart window.
exported_image	Relative image path, preferably images/sample_0_29.jpg.

3.1 Relative paths are strongly recommended

Use:

```
images/sample_0_29.jpg
```

Avoid storing a machine-specific path such as:

```
C:\Users\name\Desktop\project\dataset\images\sample_0_29.jpg
```

The program resolves a relative image path against the selected external dataset directory. This makes the same manifest usable on Windows, Google Drive, and Colab.

4. Recommended Google Drive layout

```
MyDrive/
├─ final_year_project_2_code_only.zip
├─ final_year_project_2_datasets/
│  ├─ autoscaled/
│  │  ├─ images/
│  │  └─ manifests/
│  │     ├─ train_samples.csv
│  │     └─ validation_samples.csv
│  └─ fixed_scale/
│     ├─ images/
│     └─ manifests/
│        ├─ train_samples.csv
│        └─ validation_samples.csv
└─ final_year_project_2_outputs/
```

The folder names shown above are recommendations, not hardcoded requirements. Different folder names may be used by changing the three path variables in the Colab notebook.

5. Software and hardware requirements

5.1 Python packages

Package	Role
numpy	Numerical arrays and saved embeddings.
pandas	Reading and writing manifests and reports.
Pillow	Image loading and montage creation.
matplotlib	K-selection plots.
scikit-learn	PCA, K-Means, and cluster metrics.
joblib	Saving PCA and K-Means models.
torch	CNN inference.
torchvision	Pretrained ResNet-18 and image preprocessing.

5.2 Hardware

- CPU execution is supported.
- A Colab GPU can accelerate ResNet-18 embedding extraction.
- PCA and scikit-learn K-Means normally run on the CPU.
- Sufficient storage is required for both datasets and the generated outputs.
- For large datasets, RAM usage is influenced by the number of images, embedding size, PCA components, and clustering subsamples.

5.3 First-run internet requirement

When pretrained ResNet-18 weights are enabled, the first run may download the model weights if they are not already cached. After that, embedding extraction can reuse the cached weights.

6. Understanding the configuration files

The package includes separate configuration files so the two dataset experiments remain isolated.

Configuration	Dataset	Default output
config_autoscaled.json	External auto-scaled dataset	../outputs/ autoscaled_regime_discovery
config_fixed_scale.json	External fixed-scale dataset	../outputs/ fixed_scale_regime_discovery
config.json	Any custom dataset	../outputs/custom_regime_discovery

6.1 Main configuration fields

Field	Meaning
dataset_dir	External dataset folder. May be replaced by --dataset-dir.
output_dir	Where embeddings, models, reports, plots, and montages are saved.
train_manifest	Relative path to the training manifest.
validation_manifest	Relative path to the validation manifest.
image_column	Manifest column containing the relative image path.
batch_size	Images processed together during CNN embedding extraction.
device	auto chooses CUDA when available, otherwise CPU.
pca_components	Number of retained PCA dimensions.
candidate_k	Cluster counts evaluated during model selection.
selected_k	null uses automatic recommendation; an integer forces a chosen K.
reuse_saved_embeddings	Reuses compatible saved embeddings on later runs.
overwrite_output	Deletes and rebuilds a non-empty output folder when true.

7. Running on Google Colab

The included notebook is designed for the code and datasets to remain separate in Google Drive.

7.1 Upload the required items

1. Upload final_year_project_2_code_only.zip to Google Drive.
2. Upload the auto-scaled dataset to its own Drive folder.
3. Upload the fixed-scale dataset to a different Drive folder.
4. Confirm that each dataset contains images/ and manifests/.

7.2 Open the notebook

Open run_on_colab.ipynb in Google Colab and run the Drive-mount cell.

7.3 Edit only the external path variables

```
CODE_ZIP = "/content/drive/MyDrive/final_year_project_2_code_only.zip"

AUTOSCALED_DATASET = (
    "/content/drive/MyDrive/final_year_project_2_datasets/autoscaled"
)

FIXED_SCALE_DATASET = (
    "/content/drive/MyDrive/final_year_project_2_datasets/fixed_scale"
)
```

7.4 Select the dataset experiment

For the auto-scaled experiment:

```
DATASET_TYPE = "autoscaled"
```

For the fixed-scale experiment:

```
DATASET_TYPE = "fixed_scale"
```

7.5 Select a GPU runtime

In Colab, use:

```
Runtime > Change runtime type > Hardware accelerator > GPU
```

GPU availability is not guaranteed. The package will still run on CPU when device is set to auto.

7.6 Run all notebook cells

The notebook extracts the code-only ZIP into Colab's temporary storage, installs requirements, chooses the appropriate configuration, passes the external dataset path through a command-line argument, runs the pipeline, and copies the output back to Google Drive.

7.7 Important Colab storage rule

Files stored under /content disappear when the runtime ends. Always run the final notebook cell that copies the output folder into final_year_project_2_outputs in Google Drive.

8. Running on Windows

8.1 Install dependencies

5. Extract final_year_project_2_code_only.zip.
6. Open the candlestick_regime_discovery_package folder.
7. Run install_requirements.bat.

Equivalent command:

```
python -m pip install -r requirements.txt
```

8.2 Run using Command Prompt

Auto-scaled dataset:

```
python run_unsupervised_regime_discovery.py ^
--config config_autoscaled.json ^
--dataset-dir "D:\datasets\autoscaled"
```

Fixed-scale dataset:

```
python run_unsupervised_regime_discovery.py ^
--config config_fixed_scale.json ^
--dataset-dir "D:\datasets\fixed_scale"
```

8.3 Optional output override

```
python run_unsupervised_regime_discovery.py ^
--config config_autoscaled.json ^
--dataset-dir "D:\datasets\autoscaled" ^
--output-dir "D:\results\autoscaled_run_01"
```

8.4 Environment-variable overrides

Windows Command Prompt example:

```
set FYP_DATASET_DIR=D:\datasets\autoscaled
set FYP_OUTPUT_DIR=D:\results\autoscaled_run
python run_unsupervised_regime_discovery.py --config config_autoscaled.json
```

Command-line arguments take the path decision out of the source code and allow the same package to be reused on different machines.

9. What the pipeline does

Stage	Operation
Manifest loading	Reads chronological training and validation manifests and resolves relative image paths against the selected external dataset.
CNN embedding extraction	Uses frozen ImageNet-pretrained ResNet-18 and replaces

	its classification layer with an identity mapping, producing a 512-dimensional embedding per image.
PCA fitting	Fits PCA using training embeddings only and transforms both training and validation embeddings.
Candidate K evaluation	Fits K-Means for each configured value of K and computes multiple quality and stability measures.
Automatic recommendation	Combines normalized metric scores, cluster stability, minimum-cluster proportion, and cluster balance.
Final clustering	Fits the selected K-Means model on training PCA features and assigns validation samples using <code>predict()</code> .
Inspection output	Saves representatives, nearest members, random members, and montages for each cluster.

Leakage protection: PCA and K-Means are fitted only on training images. Validation images do not influence the PCA basis or K-Means centroids.

10. Output folders and files

```
output_dir/
├─ embeddings/
├─ models/
├─ reports/
├─ plots/
└─ clusters/
```

10.1 Embeddings

- train_embeddings.npy and validation_embeddings.npy: ResNet-18 feature vectors.
- train_pca_features.npy and validation_pca_features.npy: PCA-transformed features.
- train_metadata.csv and validation_metadata.csv: manifest copies used to verify embedding compatibility.

10.2 Saved models

- pca.joblib: PCA fitted on training embeddings.
- kmeans.joblib: final K-Means model fitted on training PCA features.

10.3 Reports

File	Use
candidate_k_metrics.csv	Metrics for every evaluated K.
candidate_k_recommendation_scores.csv	Normalized component scores and combined recommendation score.
selected_k.json	Recommended K, selected K, and whether selection came from configuration or automatic recommendation.
train_cluster_labels.csv	Training labels and distance to centroid.
validation_cluster_labels.csv	Validation labels and distance to centroid.
cluster_summary.csv	Cluster sizes, proportions, and representative-sample information.
run_summary.json	Key settings and output paths for the completed run.

10.4 Cluster inspection folders

- representative.jpg: training image nearest to the cluster centroid.
- nearest_*.jpg: closest members to the centroid.
- random_*.jpg: random members for checking overall cluster consistency.
- montage_nearest.png: grid of nearest images.
- montage_random.png: grid of random images.

Nearest images show the cluster's central visual pattern. Random images reveal whether the cluster remains coherent away from the centroid.

11. How to select the number of clusters

Measure	Interpretation
Silhouette score	Higher is better. Measures within-cluster compactness and separation from other clusters.
Davies-Bouldin index	Lower is better. Measures similarity between each cluster and its most similar alternative.
Calinski-Harabasz score	Higher is better. Compares between-cluster separation with within-cluster variation.
Inertia	Lower as K increases. Use the elbow shape rather than the smallest absolute value.

Stability ARI	Higher is better. Checks whether repeated K-Means runs produce similar assignments.
Minimum cluster proportion	Avoids selecting K values that create extremely small classes.
Normalized cluster entropy	Higher indicates a more balanced cluster-size distribution.

11.1 Automatic recommendation is not the final scientific decision

The automatic score is a practical recommendation. Before accepting K, inspect every cluster's nearest and random montages. Reject a K that creates visually incoherent groups, very small clusters, or groups dominated by rendering artefacts.

11.2 Manually selecting K

After the first exploratory run, set an integer in the relevant configuration:

```
"selected_k": 6
```

Rerun the same dataset experiment. The program will fit the final model using the chosen K, provided that K is included in candidate_k.

12. Comparing auto-scaled and fixed-scale experiments

The datasets must not be mixed in a single clustering run. Run the package once for each dataset and compare the outputs.

Control	Reason
Same chronological periods	Use equivalent sample periods where possible.
Same candidate K values	Keep the cluster-search space comparable.
Same random seed	Reduces avoidable variation.
Same PCA components	Keeps feature reduction comparable.
Same CNN model and weights	Ensures differences arise mainly from chart scaling.
Separate outputs	Prevents one run from overwriting the other.

12.1 Suggested comparison table

Criterion	Auto-scaled	Fixed-scale
Recommended K		
Selected K		
Silhouette score		
Davies-Bouldin index		
Stability ARI		
Smallest cluster proportion		
Visual coherence of nearest montages		
Visual coherence of random montages		

12.2 Interpretation caution

A higher mathematical clustering score does not automatically make one scaling method economically superior. Visual coherence, cluster balance, stability, and later predictability must also be considered.

13. Safe reruns and saved embeddings

Embedding extraction is usually the most expensive stage. When `reuse_saved_embeddings` is true, the package checks whether saved embeddings and metadata remain compatible with the current manifest.

- Changing only `candidate_k` or `selected_k` can usually reuse the saved embeddings.
- Changing the dataset requires a separate output folder.
- Changing the image files or manifest ordering should trigger new embedding extraction.
- Changing the embedding model or preprocessing requires fresh embeddings.
- Set `overwrite_output` to true only when intentionally deleting the existing output folder.

For the cleanest comparison, retain separate output directories for auto-scaled and fixed-scale runs.

14. Troubleshooting

Problem	Action
Dataset directory does not exist	Check the external Drive or Windows path passed through --dataset-dir. Confirm the folder contains images and manifests.
Manifest does not exist	Confirm train_samples.csv and validation_samples.csv are inside the manifests folder.
Image does not exist	Ensure exported_image uses a valid relative path such as images/sample_0_29.jpg.
Output directory is not empty	Use a new output path, enable reuse_saved_embeddings, or intentionally set overwrite_output to true.
CUDA requested but unavailable	Use device=auto or device=cpu.
Out-of-memory during embedding extraction	Reduce batch_size from 16 to 8, 4, or 2.
Colab runtime disconnects	Use the saved-embedding reuse feature and copy outputs to Drive after each completed run.
Pretrained weights cannot download	Verify internet access on the first run or run after the weights have been cached.
K-Means takes too long	Reduce stability_subsample, silhouette_subsample, stability_repeats, or the number of candidate K values.
Clusters are dominated by one class	Try fewer clusters and inspect whether the images genuinely contain separable visual regimes.
Clusters look visually mixed	Compare alternative K values, confirm images are from the intended dataset, and inspect whether chart scaling removes or exaggerates relevant structure.

15. Recommended experimental workflow

8. Prepare and verify the auto-scaled candlestick-only dataset.
9. Prepare and verify the fixed-scale candlestick-only dataset.
10. Upload the code ZIP and both datasets separately.
11. Run the auto-scaled experiment with selected_k set to null.
12. Inspect all K-selection reports and cluster montages.
13. Choose a final auto-scaled K and rerun if necessary.
14. Run the fixed-scale experiment using the same candidate K values and other comparable settings.
15. Inspect all fixed-scale reports and montages.
16. Choose a final fixed-scale K and rerun if necessary.
17. Compare mathematical quality, stability, balance, and visual coherence.
18. Record the selected clustering model for each dataset.
19. Use the chosen labels to build future-regime prediction datasets in the next stage.

16. Final checklist

- ☐ The code-only ZIP is stored separately from the datasets.
- ☐ The auto-scaled and fixed-scale datasets are in different folders.
- ☐ Each dataset contains images/ and manifests/.
- ☐ Manifest image paths are relative.
- ☐ The appropriate dataset type is selected in Colab or the correct config is passed on Windows.
- ☐ The auto-scaled and fixed-scale runs use different output directories.
- ☐ PCA and K-Means are fitted only on training images.
- ☐ Candidate K metrics have been reviewed.
- ☐ Nearest and random montages for every cluster have been inspected.
- ☐ Clusters have not been given economic names without numerical validation.
- ☐ Outputs have been copied from Colab temporary storage to Google Drive.

- ☐ The selected K and configuration are recorded for reproducibility.

End of manual